Ramp practice solutions

Nengo ensembles, vector transformations, and recurrent dynamics

Open a solution only after attempting that question. Each green box is the result.

Part A

1 The LIF tuning curve and its two time constants

Q1

$$G(2) = \frac{1}{\tau_{\text{ref}} - \tau_{RC}\ln(1-\frac{1}{2})} = \frac{1}{0.002 + 0.02\ln 2} \approx \frac{1}{0.0159} \approx 63 \text{ Hz}.$$

As $J \to \infty$, $\ln(1 - 1/J) \to 0$, so $G \to 1/\tau_{\text{ref}} = 500$ Hz. The neuron can never fire faster than once per refractory period.

With $\tau_{\text{ref}} = 5$ ms the ceiling is $1/\tau_{\text{ref}} = 200$ Hz. A neuron asked for 200 Hz would need infinite current, and anything above it is impossible. Nengo cannot find a finite gain, so the build fails or the gain blows up. Every maximum rate must satisfy $a^{\max} < 1/\tau_{\text{ref}}$.

$G(2) = 1/(0.002 + 0.02\ln 2) \approx 63$ Hz. Ceiling $1/\tau_{\text{ref}} = 500$ Hz. At $\tau_{\text{ref}} = 5$ ms the ceiling equals the requested 200 Hz, which is unreachable.

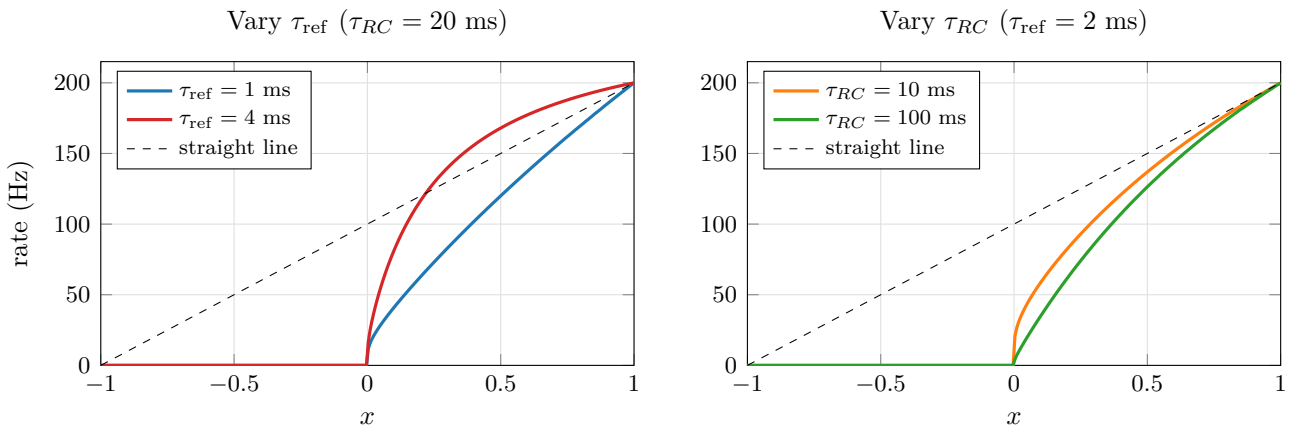
Q2 Nengo pins the curve at the intercept ($J = 1$ at $x = 0$) and at the maximum ($G = 200$ at $x = 1$). Solving $G(J^{\max}) = a^{\max}$:

$$\tau_{\text{ref}} - \tau_{RC}\ln(1 - 1/J^{\max}) = \frac{1}{a^{\max}} \quad \Longrightarrow \quad J^{\max} = \frac{1}{1 - e^{(\tau_{\text{ref}} - 1/a^{\max})/\tau_{RC}}}, \qquad J(x) = 1 + (J^{\max} - 1)\,x.$$

At 200 Hz the inter-spike interval at the maximum is 5 ms. With $\tau_{\text{ref}} = 4$ ms, only 1 ms of that is charging time, so the neuron at $x = 1$ is already pressed against its 250 Hz ceiling. Its curve shoots up just above threshold and then flattens (very concave). With $\tau_{\text{ref}} = 1$ ms there is 4 ms of charging time, the ceiling of 1000 Hz is far away, and the curve is closer to a straight line.

For τ_{RC}: the charging time $-\tau_{RC}\ln(1 - 1/J)$ behaves like τ_{RC}/J far above threshold, so the rate there is roughly linear in J. A large τ_{RC} (100 ms) needs $J^{\max} \approx 34$ to reach 200 Hz, so most of the range is far above threshold and nearly linear. A small τ_{RC} (10 ms) reaches 200 Hz already at $J^{\max} \approx 3.9$, so the whole curve sits in the sharply bending region near threshold.

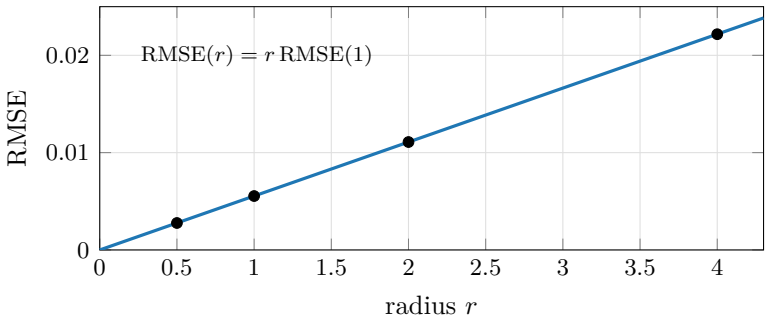
More curvature means a sum of tuning curves approximates the straight line x less well, so the RMSE is larger. Nengo, for 100 neurons: RMSE 0.0053 at $\tau_{\text{ref}} = 1$ ms and 0.0069 at 4 ms; 0.0074 at $\tau_{RC} = 10$ ms and 0.0036 at 100 ms.



Both pairs share the intercept 0 and the 200 Hz endpoint. Larger τ_{ref}: more curved, larger RMSE. Larger τ_{RC}: straighter, smaller RMSE.

2 The radius

Q1 $\text{RMSE}(r) = r \cdot \text{RMSE}(1) = 0.0055\,r$: radius $0.5 \to 0.00275$, $2 \to 0.011$, $4 \to 0.022$. Nengo gives $0.00277, 0.0111, 0.0222$. The relationship is exactly linear, because Nengo solves the same normalized problem on x/r and multiplies the answer by r.



Q2 A large radius multiplies the error on every value you actually use. If the signal only spans $[-1, 1]$, radius 4 makes the error four times larger for no benefit. Choose the radius to match the expected range.

At $x = 1.5$ with radius 1: neurons with $e = +1$ keep speeding up toward their ceilings, neurons with $e = -1$ stay silent, and the decoders, which were only fitted on $[-1, 1]$, extrapolate. The decoded value is usually somewhere between 1 and 1.5 and its error grows with distance. It is not exactly 1, because nothing in an LIF population clips hard at the radius.

RMSE $\propto r$. A radius smaller than the signal causes extrapolation error; a radius larger than needed wastes precision.

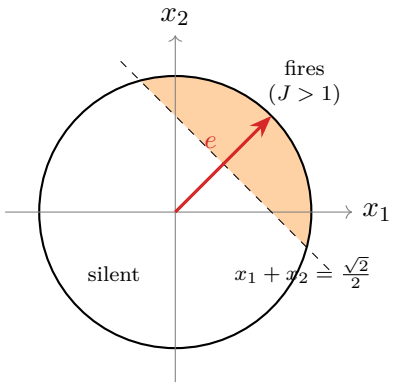
3 Encoding a 2-D vector

Q1

$$J_1 = 2(0.6) + 1 = 2.2, \qquad J_2 = 2(0.8) + 1 = 2.6, \qquad J_3 = 2(-0.36 - 0.64) + 1 = -1.$$

Neurons 1 and 2 fire ($J > 1$); neuron 3 is silent. Among unit inputs, $\langle e_3, x\rangle$ is largest for $x = e_3 = (-0.6, -0.8)$, which gives $J_3 = 3$.

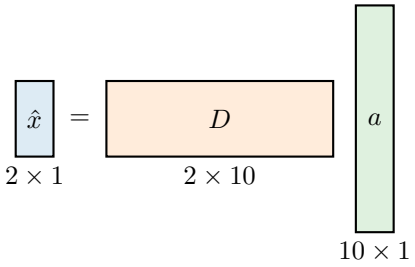
Q2 $J = \frac{x_1+x_2}{\sqrt{2}} + \frac{1}{2} > 1 \iff x_1 + x_2 > \frac{\sqrt{2}}{2} \approx 0.71$. This is the part of the disc beyond a line perpendicular to e, at distance $\frac{1}{2}$ from the origin along e.



$J = (2.2, 2.6, -1)$: neurons 1 and 2 fire. The second neuron fires for $x_1 + x_2 > \sqrt{2}/2$, a cap of the disc on the encoder side.

4 Decoding a 2-D vector, with shapes

Q1



$$D^\mathsf{T} = (AA^\mathsf{T} + N\sigma^2 I_{10})^{-1} AX^\mathsf{T},$$

with A of shape $10 \times N$, X of shape $2 \times N$, AA^T and I_{10} of shape 10×10, AX^T of shape 10×2, and D^T of shape 10×2.

Q2 $f(x) = x_1 x_2$ is one number, so its decoder is 1×10. It needs its own least-squares fit with targets $x_1 x_2$. The swap $(x_2, x_1) = Px$ with $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is linear, so its decoder is PD: the two rows of D swapped, 2×10. Linear functions of x reuse the identity decoder; nonlinear functions need a new solve.

$\hat{x} = Da$: $(2 \times 10)(10 \times 1)$. $D^{x_1 x_2}$ is 1×10. $D^{(x_2, x_1)} = PD$.

5 Linear transformations of vectors

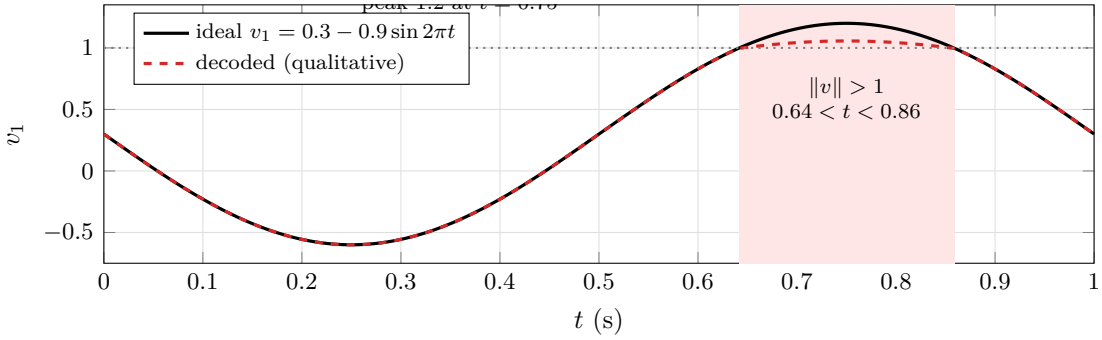
Q1

$$v = 2(0.3, -0.2) - (0.4, 0.1) + 3(-0.1, 0.2) = (0.6 - 0.4 - 0.3,\ -0.4 - 0.1 + 0.6) = (-0.1,\ 0.1),$$

so $\|v\| = \sqrt{0.02} \approx 0.14$. Connections into v: from a with `transform=2`, from b with `transform=-1`, from c with `transform=3`. Each scalar means that multiple of I_2.

Q2 $v_1(t) = 0.6 - 0.9\sin 2\pi t - 0.3 = 0.3 - 0.9\sin 2\pi t \in [-0.6,\ 1.2]$, and $v_2 = 0.1$. The largest norm is $\sqrt{1.2^2 + 0.1^2} = \sqrt{1.45} \approx 1.20$. $\|v\| > 1$ when $v_1 > \sqrt{0.99} \approx 0.995$, that is $\sin 2\pi t < -0.772$, so $t \in (0.64,\ 0.86)$ s in each second.

PN3 §1b: $w_1 = 0.5 - 3\sin 4\pi t + 0.4 - 2\sin 4\pi t = 0.9 - 5\sin 4\pi t \in [-4.1,\ 5.9]$, with $w_2 = 0.7$. That is far outside radius 1. The decoded w will be badly wrong for most of the cycle unless w gets a radius near 6.



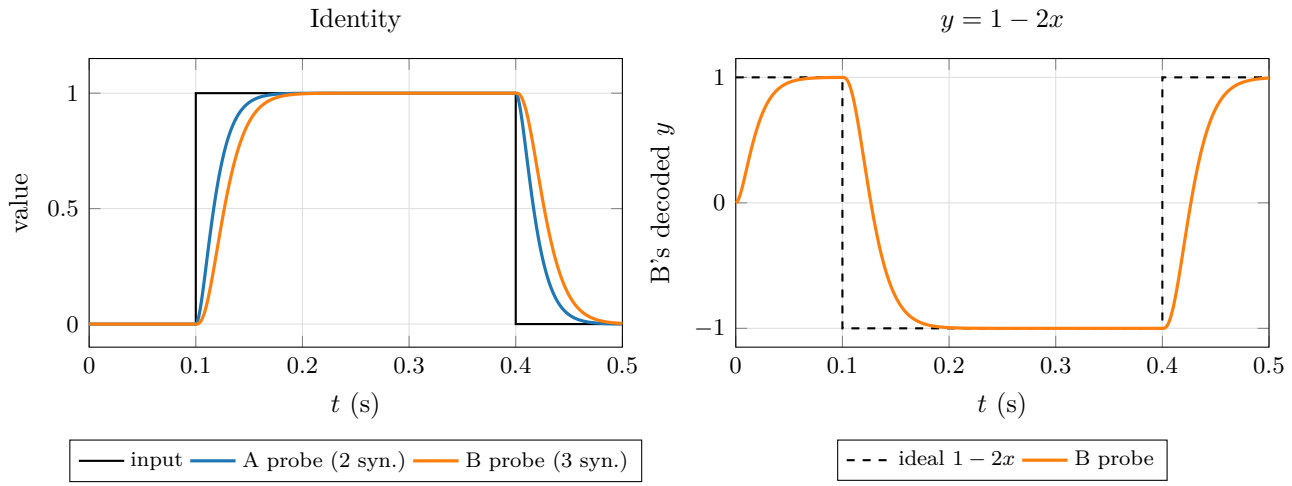
$v = (-0.1, 0.1)$, $\|v\| \approx 0.14$. With the sine: $v_1 \in [-0.6, 1.2]$, $\max \|v\| \approx 1.2$, outside the radius for $0.64 < t < 0.86$ s. PN3's $w_1 \in [-4.1, 5.9]$.

6 Synapses in a chain, and probe filters

Q1 One synapse: $x(t) = 1 - e^{-t/\tau}$, so $x(\tau) = 1 - e^{-1} \approx 0.63$ and $x(3\tau) = 1 - e^{-3} \approx 0.95$. Two synapses: $x(t) = 1 - (1 + t/\tau)e^{-t/\tau}$, so $x(\tau) = 1 - 2e^{-1} \approx 0.26$ and $x(3\tau) = 1 - 4e^{-3} \approx 0.80$. The second filter makes the start flat (zero slope at $t = 0$) and adds lag.

Q2 Input to A's probe: the input connection plus the probe, so 2 synapses. Input to B's probe: input connection, A→B, and the probe, so 3. B's trace is the most rounded and the most delayed.

For $y = 1 - 2x$, the constant 1 is decoded from A all the time, so B's probe climbs from 0 to 1 at the start of the run (two synapses: A→B and the probe). During the pulse it heads to $1 - 2 = -1$ with the three-synapse lag, and it returns to 1 after 0.4 s.



One synapse: 0.63 at τ, 0.95 at 3τ. Two: 0.26, 0.80. A's probe sits behind 2 synapses and B's behind 3. For $1-2x$, B starts by rising to 1, dips to -1 during the pulse, and returns to 1.

7 Nengo vocabulary

Q1

```
model = nengo.Network(seed=556)
with model:
    stim = nengo.Node(lambda t: 1.0 if 0.1 < t < 0.4 else 0.0)
    params = dict(intercepts=nengo.dists.Uniform(-1, 1),
                  max_rates=nengo.dists.Uniform(100, 200))
    a = nengo.Ensemble(100, dimensions=1, **params)
    b = nengo.Ensemble(50, dimensions=1, **params)
    nengo.Connection(stim, a)
    nengo.Connection(a, b, function=lambda x: 1 - 2 * x, synapse=0.01)
    p_in = nengo.Probe(stim)
    p_a = nengo.Probe(a, synapse=0.01)
    p_b = nengo.Probe(b, synapse=0.01)
with nengo.Simulator(model) as sim:
    sim.run(0.5)
```

`transform=-2` alone would give $-2x$. The offset $+1$ is not linear in x, so it needs `function=`.

Q2

(1) Line 5, input `transform=1`: it should be $B' = \tau B = \tau$, so `transform=tau`. With 1 the integral is $1/\tau = 20$ times too fast.

(2) Line 6, recurrent `transform=tau`: it should be $A' = \tau \cdot 0 + 1 = 1$. With τ the feedback is $0.05x$, a strong leak rather than memory.

(3) Line 6, recurrent `synapse=0.005`: A' and B' were derived for the recurrent synapse, so it must be `synapse=tau` (50 ms).

Input: `transform=tau, synapse=0.005`. Recurrent: `transform=1, synapse=tau`.

Part B

8 Two Laplace facts

Q1

$$H(s) = \int_0^\infty \frac{1}{\tau} e^{-t/\tau} e^{-st}\, dt = \frac{1}{\tau}\int_0^\infty e^{-t(s+1/\tau)} dt = \frac{1}{\tau}\left[\frac{-e^{-t(s+1/\tau)}}{s+1/\tau}\right]_0^\infty = \frac{1}{\tau}\cdot\frac{1}{s+1/\tau} = \frac{1}{1+s\tau}.$$

The upper limit is 0 because $\mathrm{Re}(s) > -1/\tau$; the lower limit gives $-1/(s+1/\tau)$, and the minus signs cancel.

Q2 With $a = s + 1/\tau$, integration by parts gives $\int_0^\infty te^{-at}dt = \left[-\frac{t}{a}e^{-at}\right]_0^\infty + \frac{1}{a}\int_0^\infty e^{-at}dt = \frac{1}{a^2}$. So

$$\mathcal{L}\{h_1\} = \frac{1}{\tau^2(s+1/\tau)^2} = \frac{1}{(1+s\tau)^2} = H(s)^2.$$

h_1 is two exponential synapses in series, which is why its step response matched the two-synapse curve in Section 6.

Derivative check: $x = e^{-at}$ has $X = 1/(s+a)$. Directly, $\mathcal{L}\{-ae^{-at}\} = -a/(s+a)$. The rule gives $sX - x(0) = \frac{s}{s+a} - 1 = \frac{-a}{s+a}$. They agree. Lecture 6 drops $x(0)$ because the signals start at zero.

> $H(s) = 1/(1+s\tau)$. $\mathcal{L}\{h_1\} = 1/(1+s\tau)^2$. $\mathcal{L}\{\dot{x}\} = sX - x(0)$.

9 The recurrent connection is a differential equation

Q1

$x = h * (g(u) + f(x))$	the population represents its filtered total input
$X(s) = \frac{1}{1+s\tau}\big(G(s) + F(s)\big)$	convolution becomes multiplication by $H(s)$
$X(1+s\tau) = G + F$	multiply through
$s\tau X = G + F - X$	isolate the s term
$\tau\dot{x} = g(u) + f(x) - x$	s becomes d/dt

Q2 $y(t) = \int_{-\infty}^t \frac{1}{\tau} e^{-(t-t')/\tau} z(t')\, dt'$. Differentiating the upper limit gives $\frac{1}{\tau}z(t)$, and differentiating the integrand gives $-\frac{1}{\tau}y(t)$. So $\dot{y} = (z-y)/\tau$, which is $\tau\dot{y} = z - y$. Substituting $z = g(u) + f(x)$ and $y = x$ gives the same equation.

The synapse integrates its input (the z/τ term) but loses its own value at rate y/τ (the leak). Left alone it forgets with time constant τ. That is a leaky integrator.

> $\tau\dot{x} = g(u) + f(x) - x$. The $-x$ is the synaptic leak.

10 Linear feedback $f(x) = kx$

Q1 $\tau\dot{x} = u - 2x$, so $\dot{x} = \frac{2}{\tau}\left(\frac{u}{2} - x\right)$ and $x(t) = \frac{1}{2}(1 - e^{-2t/\tau})$. The steady state is $\frac{1}{2}$ and the time

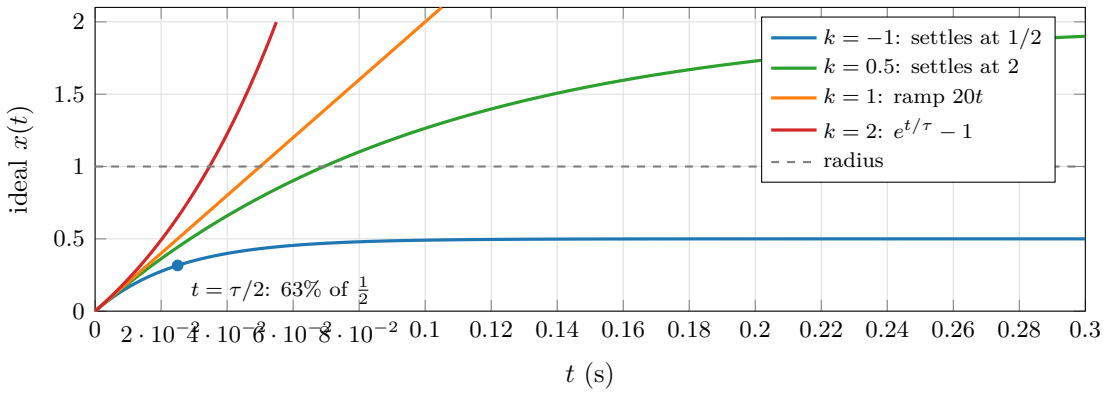
constant is $\tau/2 = 25$ ms. The lecture's "converges to $u/2$" is the steady state. The equation also says the loop approaches it twice as fast as the synapse alone.

Q2

- $k = 0.5$: $\tau\dot{x} = u - 0.5x$, so $x = 2(1 - e^{-t/(2\tau)})$. The steady state 2 is outside radius 1, so the neural trace bends over near 1. Time constant $2\tau = 100$ ms.
- $k = 1$: $\tau\dot{x} = u$, so $x = t/\tau = 20t$, which reaches the radius at $t = 50$ ms. An integrator.
- $k = 2$: $\tau\dot{x} = u + x$, so $x = e^{t/\tau} - 1$, which reaches 1 at $t = \tau \ln 2 \approx 35$ ms. Unstable.

Lecture's $f(x) = x + 1$: $\tau\dot{x} = u + x + 1 - x = u + 1$. The x terms cancel, so it is an integrator of $u + 1$. With $u = 0$ it ramps at $1/\tau$ forever. The "convergence slightly above 1" in the lecture is saturation, not a real equilibrium.

Unit step, $\tau = 50$ ms (ideal curves; neural traces flatten near the radius)



$k = -1$: $\frac{1}{2}(1 - e^{-2t/\tau})$. $k = 0.5$: $2(1 - e^{-t/2\tau})$. $k = 1$: t/τ. $k = 2$: $e^{t/\tau} - 1$. $f = x + 1$ is an integrator of $u + 1$.

11 One-dimensional phase portraits and stability

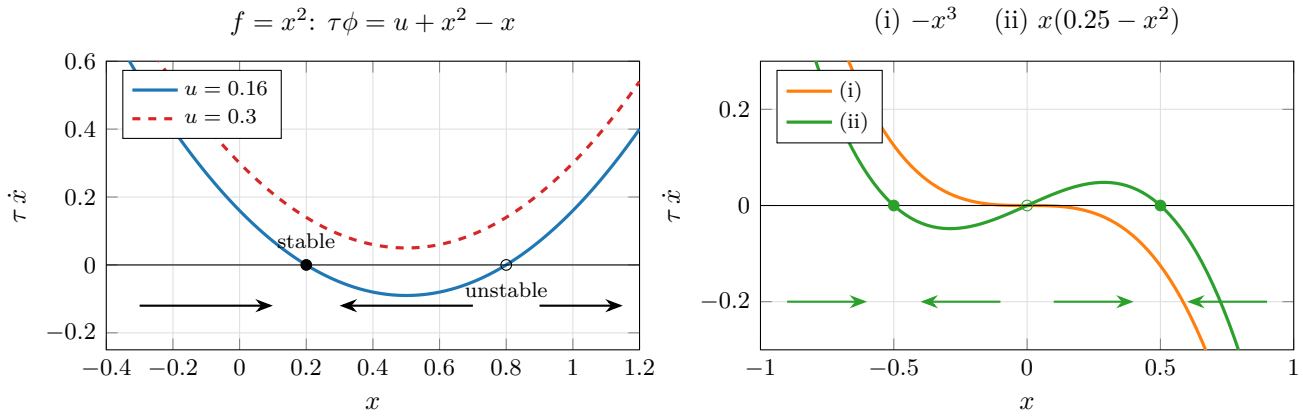
Q1 $\tau\phi = u + x^2 - x$. At $u = 0.16$: $x^2 - x + 0.16 = 0$, so $x = \frac{1 \pm \sqrt{1 - 0.64}}{2} = \frac{1 \pm 0.6}{2} = 0.2$ or 0.8. With $\tau\phi' = 2x - 1$: at 0.2 it is -0.6 (stable), and at 0.8 it is $+0.6$ (unstable).

At $u = 0.3$ the discriminant $1 - 1.2$ is negative, so there is no equilibrium and $\phi > 0$ everywhere: x grows until it saturates. Ideally the jump happens at $u = 1/4$. The lecture saw about 0.4 because the neural version is distorted and saturates.

With $x(0) = 1$ and $u = 0.16$ the state starts right of the unstable point, so it runs away upward and never returns to 0.2. The same input gives different outcomes depending on where the state was. That history dependence is the lecture's "memory".

Q2 (i) $\tau\phi = -x^3$. The only equilibrium is $x = 0$. Here $\phi'(0) = 0$, but the arrows point inward on both sides, so it is stable. Convergence is slow and not exponential.

(ii) $\tau\phi = x(0.25 - x^2)$, with zeros at 0 and ± 0.5. $\tau\phi' = 0.25 - 3x^2$: $+0.25$ at 0 (unstable), -0.5 at ± 0.5 (stable). (ii) stores one bit, the sign of x: any start with $x > 0$ settles at $+0.5$, and any start with $x < 0$ settles at -0.5.



Filled dots are stable and open dots unstable. Arrows show the direction of motion along x.

$u = 0.16$: stable 0.2, unstable 0.8; no equilibrium for $u > 1/4$. (i) one slow stable point at 0. (ii) stable ± 0.5, unstable 0; it stores the sign of x.

12 Mapping an LTI system onto a synapse

Q1 $X = \frac{1}{1+s\tau}(A'X + B'U) \Rightarrow s\tau X = (A' - I)X + B'U \Rightarrow sX = \frac{A'-I}{\tau}X + \frac{B'}{\tau}U$. Matching with $sX = AX + BU$ gives $A = (A' - I)/\tau$ and $B = B'/\tau$, so $A' = \tau A + I$ and $B' = \tau B$.

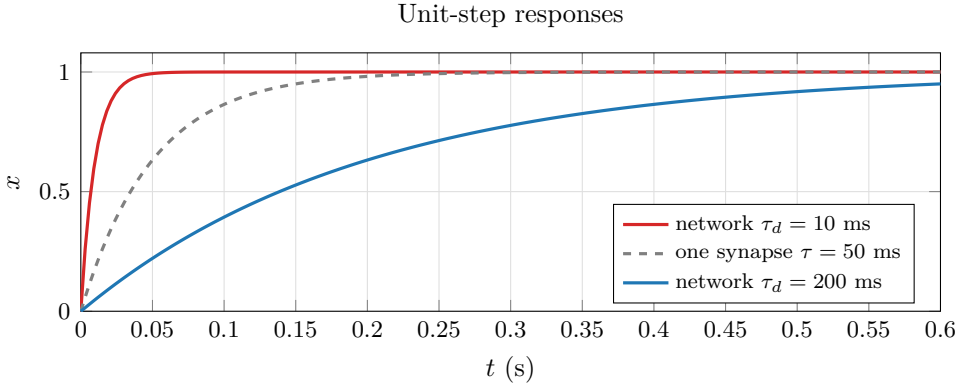
Integrator: $A = 0$, $B = I_d$, so $A' = I_d$ and $B' = 0.05\, I_d$.

Q2 $A = -1/\tau_d$, $B = 1/\tau_d$.

$$\tau_d = 200 \text{ ms}: \; A' = 1 - \tfrac{0.05}{0.2} = 0.75, \; B' = 0.25. \qquad \tau_d = 10 \text{ ms}: \; A' = 1 - \tfrac{0.05}{0.01} = -4, \; B' = 5.$$

Steady state $x = A'x + B'u$: $0.25x = 0.25u$ and $5x = 5u$, so both give $x = u$.

The surprise: with $\tau_d = 10$ ms the network responds five times faster than its own 50 ms synapses. Strong negative feedback ($A' = -4$) combined with a large input gain speeds the loop up. The network-level time constant is a design choice, separate from the synapse's.

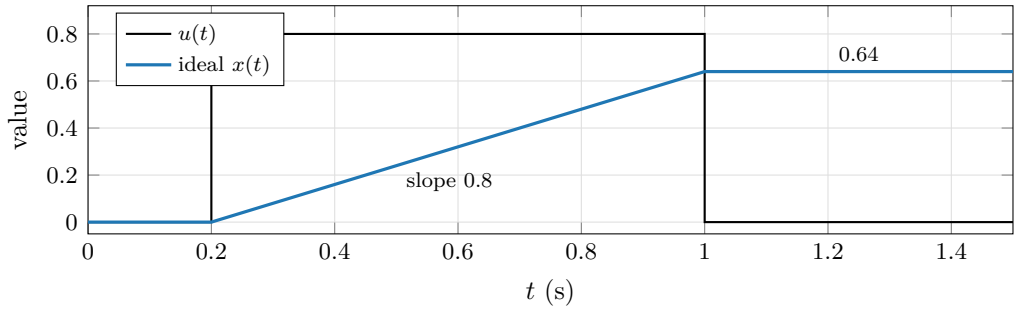


$A' = \tau A + I$, $B' = \tau B$. Integrator: $A' = I$, $B' = 0.05I$. Low-pass: $(0.75, 0.25)$ for 200 ms, $(-4, 5)$ for 10 ms.

13 The ideal answer is a definite integral

Q1

$$x(t) = \int_0^t u = \begin{cases} 0, & t < 0.2, \\ 0.8(t - 0.2), & 0.2 \le t < 1.0, \\ 0.64, & t \ge 1.0. \end{cases}$$

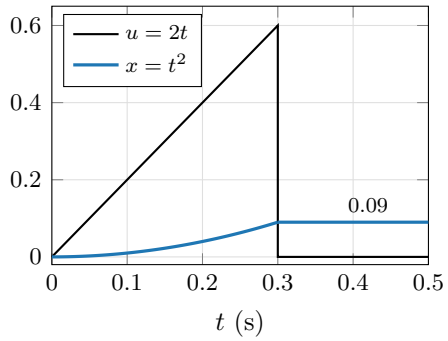


Q2 (i) The ramp is $u = 2t$ on $[0, 0.3)$, so $x = t^2$ there and then holds at 0.09.

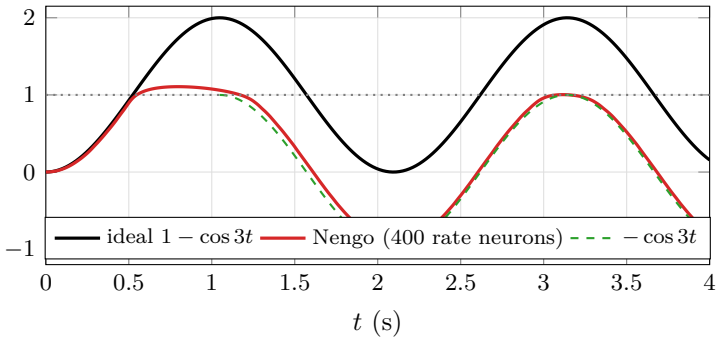
(ii) $x = \int_0^t 3 \sin 3t' \, dt' = [-\cos 3t']_0^t = 1 - \cos 3t$. Its range is $[0, 2]$, which exceeds radius 1.

The neural state saturates near 1 at $t = \pi/6 \approx 0.52$ s, while the ideal keeps climbing to 2 at $t = \pi/3$. The integrator only stores its current value, so when u goes negative it comes down by the full integral and ends up 1 below the ideal. From then on it traces $1 - \cos 3t - 1 = -\cos 3t$, which fits inside the radius. The simulated 400-neuron trace does exactly this. PN3's $5 \sin 5t$ (ideal $1 - \cos 5t$) has the same problem.

(i) ramp



(ii) $u = 3 \sin 3t$, radius 1



Step: 0, $0.8(t - 0.2)$, 0.64. Ramp: t^2, then 0.09. Sine: $1 - \cos 3t \in [0, 2]$; the neural trace clips once, then follows about $-\cos 3t$.

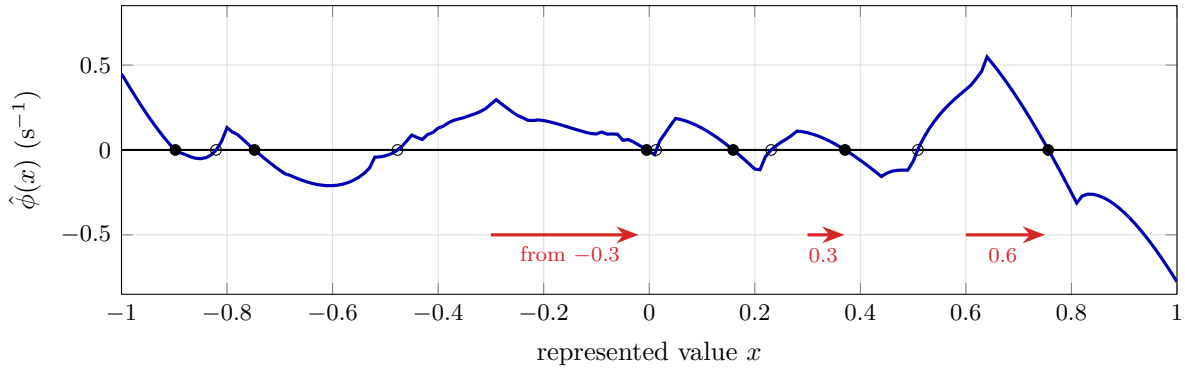
14 Why a neural integrator drifts

Q1 Read the sign of $\hat{\phi}$ at the start, then follow it to the next zero where $\hat{\phi}$ crosses downward.

- $x = 0.3$: $\hat{\phi} \approx +0.1$ s^{-1}, so it moves up to the stable zero near 0.37. It covers 0.07 at under 0.15 per second, slowing as it arrives: on the order of a second.

- $x = -0.3$: $\hat{\phi} \approx +0.27$, so it moves up to the stable zero near 0. This takes one to two seconds.
- $x = 0.6$: $\hat{\phi} \approx +0.36$, so it moves up to the stable zero near 0.76, in well under a second.

The stable zeros (downward crossings) are near $-0.90, -0.75, 0, 0.16, 0.37, 0.76$. The unstable ones are near $-0.82, -0.48, 0.01, 0.23, 0.51$.



Q2 In spiking mode, each run adds a different realisation of spike noise to the recurrent input. The state takes a random walk on top of the deterministic drift, so it can be knocked across an unstable zero into a different basin, and repeated runs end in different places. In rate mode there is no spike noise, the dynamics are exactly $\hat{\phi}$, and the same seed gives the same neurons, so the run repeats.

Doubling the neurons roughly halves the distortion error. The bumps in $\hat{\phi}$ get smaller (slower drift) and usually more numerous and closer together. The curve never becomes exactly zero.

Drift to the next stable zero: $0.3 \to 0.37$, $-0.3 \to 0$, $0.6 \to 0.76$. Static distortion causes deterministic drift; spike noise causes run-to-run variation.

15 Unequal input and recurrent synapses

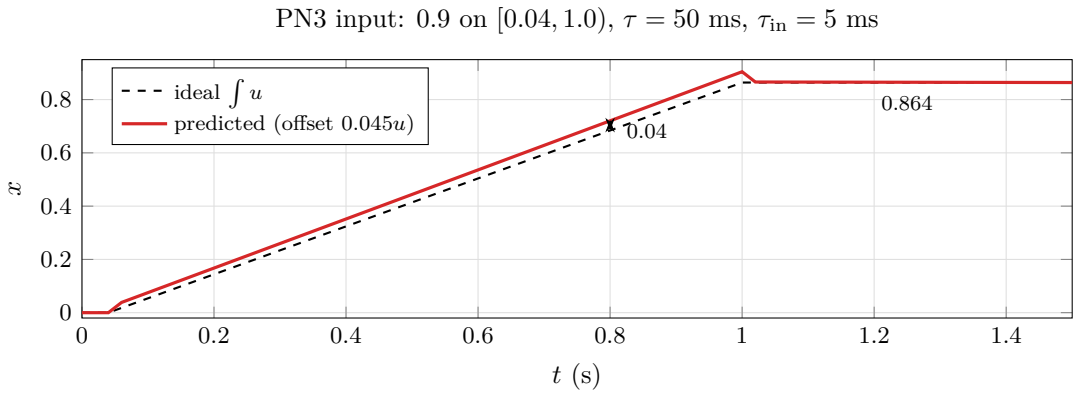
Q1 With $A' = 1$ and $B' = \tau$:

$$X = \frac{X}{1+s\tau} + \frac{\tau U}{1+s\tau_{\text{in}}} \;\Rightarrow\; X\,\frac{s\tau}{1+s\tau} = \frac{\tau U}{1+s\tau_{\text{in}}} \;\Rightarrow\; X = \frac{1+s\tau}{1+s\tau_{\text{in}}} \cdot \frac{U}{s} = \frac{U}{s} + \frac{(\tau - \tau_{\text{in}})}{1+s\tau_{\text{in}}} U.$$

The first term is the ideal integral. The second is the input itself, filtered by the short synapse and weighted by $\tau - \tau_{\text{in}}$.

Q2 Offset while on: $(\tau - \tau_{\text{in}})u = 0.045 \times 0.9 = 0.0405$. The trace jumps up by about 0.04 within a few ms of $t = 0.04$, then rises parallel to the ideal. At $t = 1.0$ it drops by 0.04 onto the ideal final value $0.9 \times 0.96 = 0.864$. The end value is unaffected.

If B' were computed with τ_{in}, the input gain would be 0.005 instead of 0.05, so the network would integrate at one tenth of the rate (final value about 0.086).



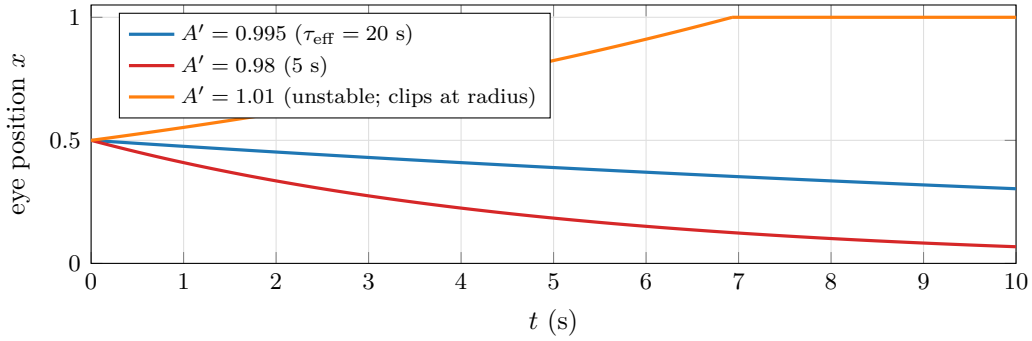
$x = \int u + (\tau - \tau_{\text{in}})(h_{\tau_{\text{in}}} * u)$. Offset 0.0405 while the input is on; final value 0.864 as intended.

16 A leaky integrator: eye position

Q1 $A' = 1 - 0.1/20 = 0.995$ and $B' = 0.1$. With $v = 0$: $x(t) = 0.5e^{-t/20}$, so $x(20) = 0.5/e \approx 0.18$.

Q2 $A = (A' - 1)/\tau = -0.02/0.1 = -0.2\ \text{s}^{-1}$, so $\tau_{\text{eff}} = 5$ s and $x = 0.5e^{-t/5}$: $0.5e^{-2} \approx 0.07$ at 10 s, against $0.5e^{-0.5} \approx 0.30$ intended. With $A' = 1.01$: $A = +0.1$, so $x = 0.5e^{t/10}$ grows until it saturates.

A decoding error of 1.5% in the recurrent gain cut the memory from 20 s to 5 s, and 1% the other way makes it unstable, because $\tau_{\text{eff}} = \tau/(1 - A')$ divides by a tiny number. Long memories need very accurate recurrent decoders, so many neurons or some correction.



$A' = 0.995$, $B' = 0.1$; $x(20) \approx 0.18$. $A' = 0.98 \Rightarrow \tau_{\text{eff}} = 5$ s. $A' > 1$ is unstable.

17 The oscillator

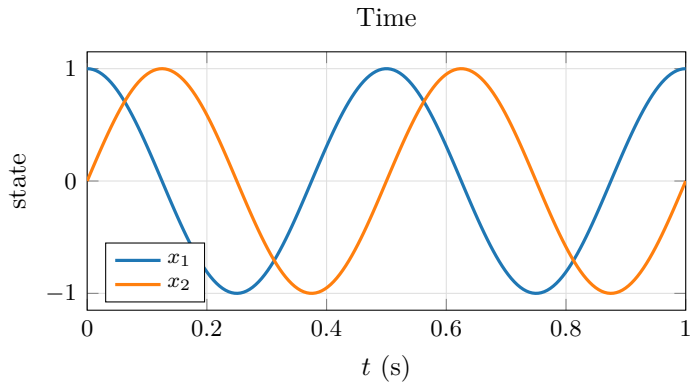
Q1 $\omega = 2\pi \cdot 2 = 4\pi$, so $\tau\omega = 0.4\pi \approx 1.26$ and

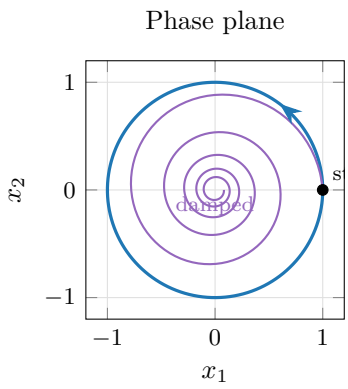
$$A' = \begin{pmatrix} 1 & -0.4\pi \\ 0.4\pi & 1 \end{pmatrix}, \qquad B' = 0.$$

From $(1, 0)$: $x_1 = \cos 4\pi t$ and $x_2 = \sin 4\pi t$. The trajectory is a counter-clockwise circle of radius 1, once every 0.5 s.

Q2 At $x = 0$, $Ax = 0$, so the origin is an equilibrium and nothing moves without a kick. A huge kick cannot give a huge oscillation, because the neurons saturate at the edge of the represented range. The amplitude settles near the radius, which the lecture describes as a stable cyclic attractor. Noise and distortion can also make a neural oscillator decay to zero.

Damped: $A' = \tau A + I = \begin{pmatrix} 1-\tau\gamma & -\tau\omega \\ \tau\omega & 1-\tau\gamma \end{pmatrix}$, with trajectory $e^{-\gamma t}(\cos\omega t, \sin\omega t)$, an inward spiral.





$A' = \begin{pmatrix} 1 & -0.4\pi \\ 0.4\pi & 1 \end{pmatrix}$, $B' = 0$. Circle of period 0.5 s, counter-clockwise; it needs a kick, and its amplitude is set by the radius.

18 Nonlinear dynamics with the same recipe

Q1 $f'(x) = \tau(-x^3) + x = x - 0.1x^3$ and $g'(u) = 0.1u$.

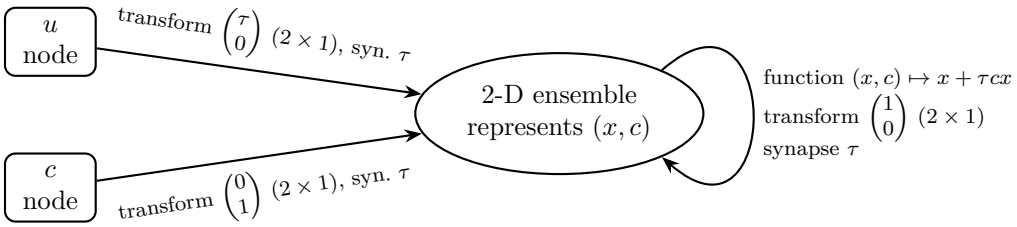
```
nengo.Connection(stim, x, transform=0.1, synapse=0.1)
nengo.Connection(x, x, function=lambda x: x - 0.1 * x**3, synapse=0.1)
```

Q2 $f'(x) = \tau \begin{pmatrix} 10x_2 - 10x_1 \\ -x_1x_3 - x_2 \\ x_1x_2 - \frac{8}{3}(x_3 + 28) - 28 \end{pmatrix} + x$. The products x_1x_3 and x_1x_2 are nonlinear in more than one variable. As with xy in Test 2, they can only be decoded from one population whose neurons are tuned to all three variables together. No kick is needed because the origin is not an equilibrium: the third component is $-\frac{8}{3} \cdot 28 - 28 \neq 0$ there, so the state starts moving on its own.

$f'(x) = x - 0.1x^3$, $g' = 0.1u$. Lorenz: one 3-D population, recurrent function $\tau f(x) + x$, no kick needed.

19 The controlled integrator

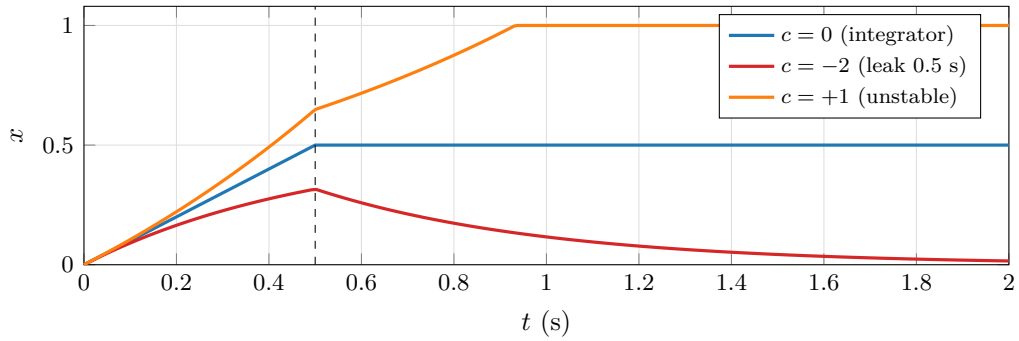
Q1



Dimension 2 has no recurrence, so it just follows the filtered c. Only dimension 1 is integrated.

Q2

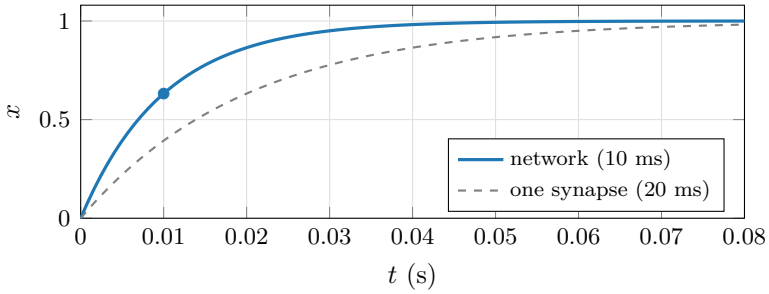
- $c = 0$: $x = t$ up to 0.5, then it holds at 0.5.
- $c = -2$: $\dot{x} = -2x + 1$, so $x = \frac{1}{2}(1 - e^{-2t})$, which is $\frac{1}{2}(1 - e^{-1}) \approx 0.32$ at $t = 0.5$. After that it decays as $0.32e^{-2(t-0.5)}$, with time constant $1/|c| = 0.5$ s.
- $c = +1$: $\dot{x} = x + 1$, so $x = e^t - 1 \approx 0.65$ at $t = 0.5$. Then $0.65e^{t-0.5}$ reaches 1 at $t \approx 0.93$ s and saturates.



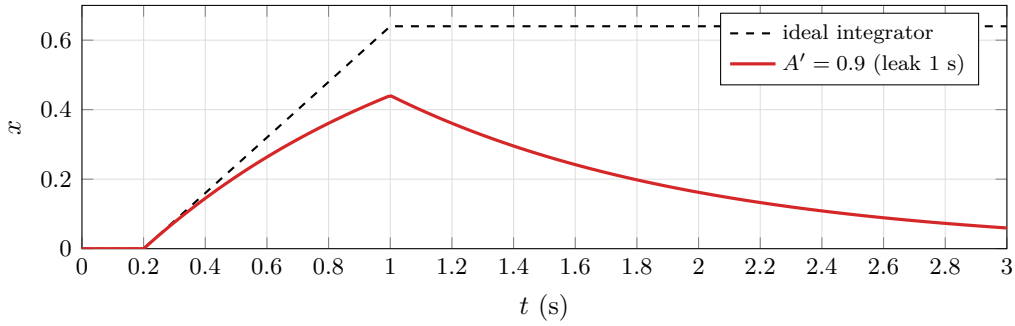
u in with $(\tau, 0)^\mathsf{T}$, c in with $(0, 1)^\mathsf{T}$, recurrent $x + \tau cx$ into dimension 1. $c = 0$ holds 0.5; $c = -2$ peaks at 0.32 and decays with 0.5 s; $c = +1$ runs to the radius.

Part C

Q1 $\tau\dot{x} = 2u - x - x = 2(u - x)$, so $\dot{x} = (u - x)/(\tau/2)$. This is a low-pass filter with steady state u and time constant $\tau/2 = 10$ ms: $x = 1 - e^{-t/0.01}$, reaching 63% at 10 ms.



Q2 $A = (0.9-1)/0.1 = -1\ \text{s}^{-1}$ and $B = B'/\tau = 1$, so $\dot{x} = -x + u$: a leaky integrator with a 1 s time constant. For $u = 0.8$ on $[0.2, 1.0)$: $x = 0.8(1 - e^{-(t-0.2)})$, which is $0.8(1 - e^{-0.8}) \approx 0.44$ at $t = 1$, then $0.44e^{-(t-1)}$. The ideal holds 0.64.



Q3 A synapse does not hold its value: it drifts toward whatever input it is given, losing its own value over a time of about τ. Feeding the represented value back as input (the "$+x$") cancels that loss, so the loop holds its state and only changes when we add something extra. Multiplying the desired rate of change by τ converts it into the size of input the slow synapse needs in order to move at exactly that rate.